Mark schemes

Q1.

$$\bar{X} = \frac{2 \times 9 + 3 \times 2 + 8 \times 3 + 7 \times 6}{2 + 3 + 8 + 7}$$

M1: Expression for $\bar{X}$ with no more than one error in the numerator and correct denominator.

M1

$= \frac{90}{20}$ or 4.5

A1: Correct distance.

Accept $\frac{9}{2}$ or $\frac{90}{20}$ or equivalent.

A1

$$\bar{Y} = \frac{2 \times 6 + 3 \times 4 + 8 \times 8 + 7 \times 11}{20}$$

M1: Expression for $\bar{Y}$ with no more than one error in the numerator and correct denominator.

M1

$= \frac{165}{20}$ or 8.25

A1: Correct distance.

Accept $\frac{33}{4}$ or $\frac{165}{20}$ or equivalent

A1

∴ Centre of mass is at (4.5, 8.25)

A1: Correct coordinates; dependent on M1 M1

Do not accept $\frac{90}{20}$ etc at this stage.

SC4: For final answer (8.25, 4.5) award 4 marks.

Moments about B, (2.5, 4.25) SC2

A1F

[5]

Q2.

$$\bar{X} = \frac{3 \times 15 + 1 \times 7 + 6 \times 8 + 10 \times 12}{3 + 1 + 6 + 10}$$

M1 for at least 3 multiplication & addition

M1A1

$= \frac{220}{20}$ or 11

A1

$\bar{Y} = \frac{3 \times 6 + 1 \times 14 + 6 \times 7 + 10 \times 9}{20}$

M1A1

$= \frac{164}{20}$ or 8.2

SC 4 (10, 7.4) [omit lamina] ie: B2, B2

A1

∴ Centre of mass is at (11, 8.2)

6

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Q3.

$\bar{X} = \frac{25 \times 1 + 12 \times 4 + 4 \times 5}{1 + 4 + 5}$

Two terms on top correct (+ third) and denominator correct

M1

$= \frac{93}{10}$ or 9.3

A1

$\bar{Y} = \frac{10 \times 1 + 7 \times 4 + 18 \times 5}{10}$

M1

$= \frac{128}{10}$ or 12.8

SC3 interchange $\bar{X}$ *and* $\bar{Y}$

A1

∴ Centre of mass is at (9.3, 12.8)

4

[4]

Q4.

(a) $\bar{x} = \dfrac{4 \times 90 + 7 \times 90}{4 + 7 + 8 + 11}$

moment equation

M1

correct equation

A1

$= \dfrac{990}{30} = 33 \text{ cm}$ AG

correct distance from correct working

A1

3

(b) $\bar{y} = \dfrac{11 \times 60 + 7 \times 60}{30}$

moment equation

M1

correct equation

A1

$= \dfrac{1080}{30}$ = 36 cm

correct distance

A1

3

(c) tan $\alpha = \dfrac{36}{33}$

use of tan

correct expression

M1 A1F

α = 47.5°

correct angle

follow through $\bar{y}$ *from part (b).*

A1F

3

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Q5.

(a) 5cm

B1

1

(b) $\Sigma mx = (\Sigma m)\bar{x}$

M1 one side correct

About PS:
$6(m) + 6(m) + 3(1) = (2m + 1)\bar{x}$

M1A1

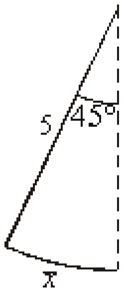
$$\bar{x} = \frac{12m + 3}{2m + 1}$$

AG

A1

3

(c)



$\tan 45° = \frac{\bar{x}}{5}$

Principle applied
Equation correct – use of part (a)

M1A1

$$\Rightarrow \frac{12m + 3}{2m + 1} = 5$$

Substitute and tan 45° = 1

A1

$12m + 3 = 10m + 5$

Solving - dependent

m1

$2m = 2$
$m = 1$

CAO

A1

5

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Q6.

(a) Σmx $\Sigma m\bar{x}$

$2(1) + m(7) = (2 + m)5$

mx correct for one term

M1

Fully correct equation

A1

$\therefore$ $m = 4$

m = 4 obtained correctly; AG

A1

3

***Alternative:** moments about x = 5*

***(a)** 2(4) = m (2)*

M1A1

m = 4

A1 (3)

(b) Σmx $\Sigma m\bar{y}$

my correct for one term
(m need not be substituted)

M1

$2(8) + 4(11) = 6\bar{y}$

Fully correct equation

A1

$\bar{y} = 10$

$\bar{y}$ *correctly obtained from their equation*
(allow one slip)

A1

3

***Alternative**: moments about y =* $\bar{y}$

(b)2 ($\bar{y}$ *– 8) = 4(11 –* $\bar{y}$*)*

M1A1

$\bar{y}$ *= 10*

A1

[6]

Q7.

(a) $\bar{x} = \dfrac{3 \times 1 + 12 \times 2.5 + 15 \times 3 + 10 \times 2}{3 + 12 + 15 + 10}$

Five term moment equation

M1

Correct numerator

A1

$= \dfrac{98}{40}$ = 2.45 m

Correct denominator

A1

ag *Correct answer from correct working*

A1

4

(b) $4T_B = 2.45 \times 40 \times 9.8$

Two term moment equation

M1

T_B = 240 N (to 3 sf)

Correct moment equation

A1

Correct tension

A1

$T_A + 240 = 40 \times 9.8$

Vertical equilibrium or second moment equation

m1

T_A = 152 N (to 3 sf)

Correct second tension

A1

5

[9]

Q8.

(a) $x = \dfrac{0.6 \times 8 + 1.2 \times 4 + 1.2 \times 4}{20} = 0.72\,\text{m}$

Equation with 3 non-zero terms on RHS

M1

Correction equation

A1

Correct value

A1

3

(b) Due to symmetry of masses in line through centre parallel to AD

Valid explanation or calculation

B1

1

(c) $\tan \alpha = \frac{0.72}{0.6}$

Use of tan

M1

Use of $\bar{x}$ and $\bar{y}$

M1

Correct equation

A1 ft

$\alpha = 50.2°$

Correct angle

A1 ft.

4

[8]

Examiner reports

Q1.

This question was answered well by virtually all candidates.

Q2.

This question was also answered well by most candidates. However, a significant number of candidates ignored the lamina in their solutions. A small number of candidates made arithmetical errors.

Q3.

The majority of candidates answered this question well, although some divided all the masses by two or tried to consider a lamina. A few candidates interchanged $\overline{X}$ and $\overline{Y}$.

Q4.

Parts (a) and (b) of this question were done well by many candidates. They were probably helped by the printed answer in part (a) to select the correct method and were able to apply this again in part (b). Some candidates did use some unconventional methods, such as working relative to a point outside the framework, to obtain correct solutions.

Part (c) was found to be more demanding. Candidates made a number of errors including;

- Using sine or cosine instead of tangent functions,
- Not using the correct distances,
- Using the correct distances, but with the fraction inverted.

Q5.

Very few marks were dropped in part (a), where sometimes an answer of 3 cm was seen. Marks were sometimes lost in part (b) because candidates failed to realise exactly how to use $\Sigma(mx) = (\Sigma m)x$ to obtain the printed result in part (b).

In the final part, some candidates failed to identify the correct right angled triangle or made errors with basic trigonometry. Some candidates did not appear to be aware that the triangle only involved using the tangent ratio. Some incorrect cancelling was seen, which resulted in the wrong answer being obtained for m.

Q6.

This proved to be a sound starter question. Many candidates clearly knew the relevant formulae well. The standard method of using $\sum mx$ proved most successful. Where candidates attempted other methods such as ratios, equations of lines etc, these were less successful.

Q7.

This question caused more difficulty for a fair number of candidates. Quite a lot managed to produce correctly the printed answer for part (a).

Part (b) caused much more difficulty. Some candidates did not attempt this or did not get beyond a statement like $T_1 + T_2 = 392$. Some did produce quite a bit of incorrect working. There were some candidates who did not multiply the unknown tension by a distance, although they did do this for the weight of each particle and of the rod. Interestingly quite a few candidates did not use the answer from part (a), but treated each particle separately. Another common error was to omit g from the calculations.

Q8.

There were some very good responses to this question, but some candidates did find parts (a) and (b) quite difficult. In part (a), the common error was to omit the 8 kg particle from the calculations. The candidates who did include the 8 kg particle normally obtained the correct answer, but some did make arithmetic errors. There were some good answers to part (b), but these were not very common. Some candidates did not make it clear that the distribution of the masses was symmetrical and simply commented on the square being symmetrical. A few candidates carried out a calculation to explain the result. There were some good responses to part (c), with several candidates taking the correct approach.